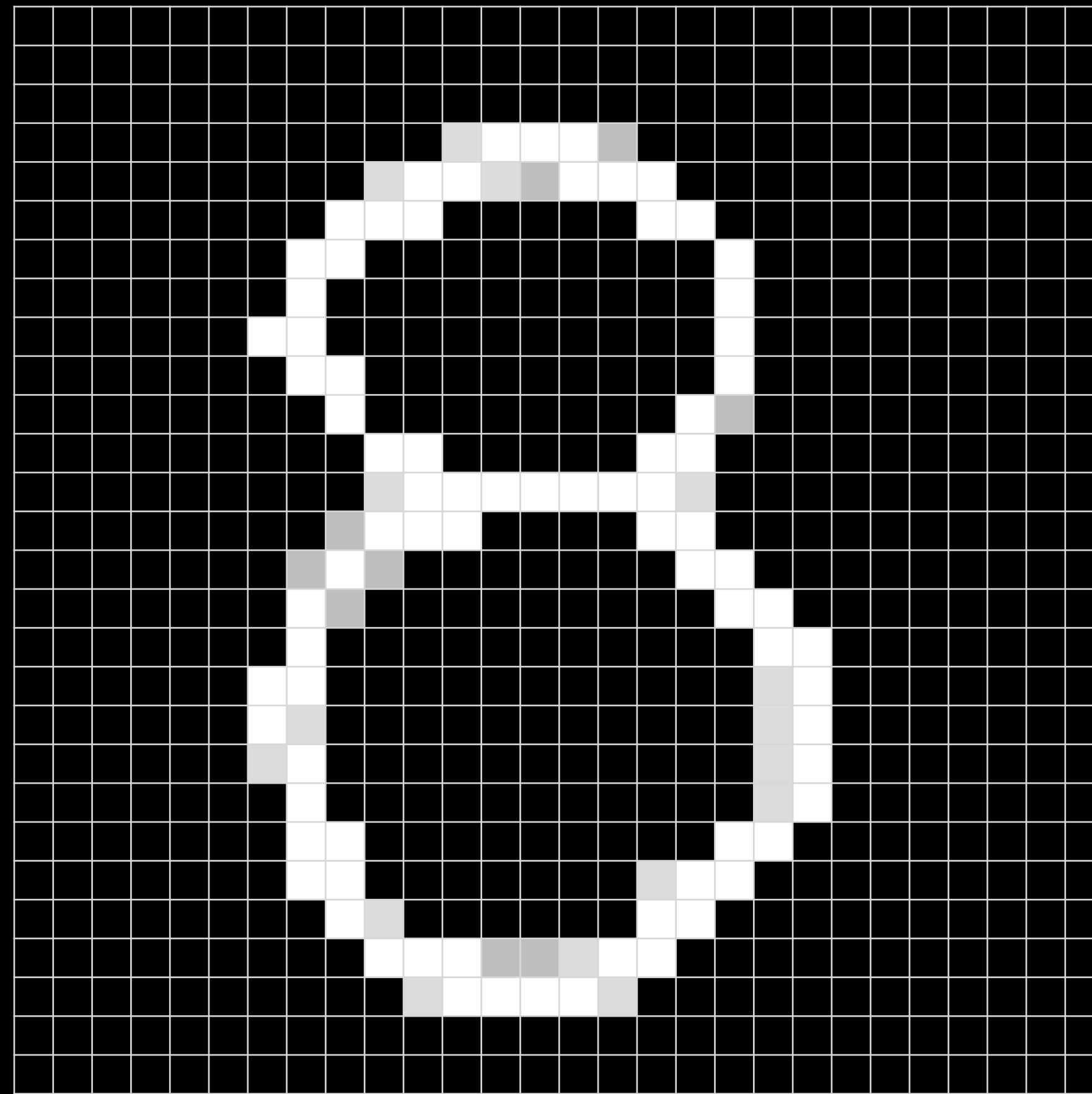
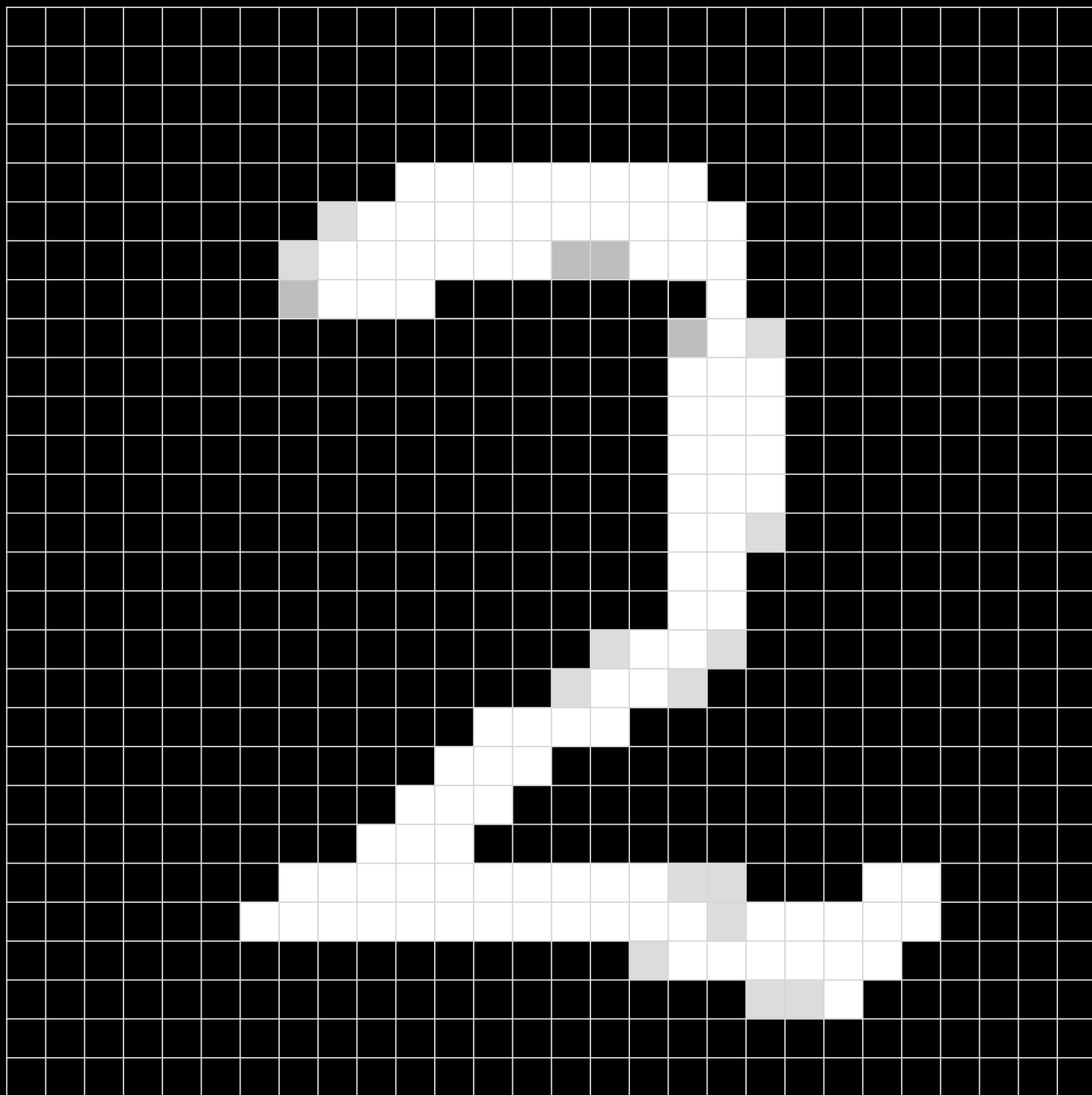
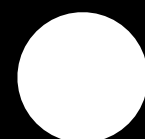
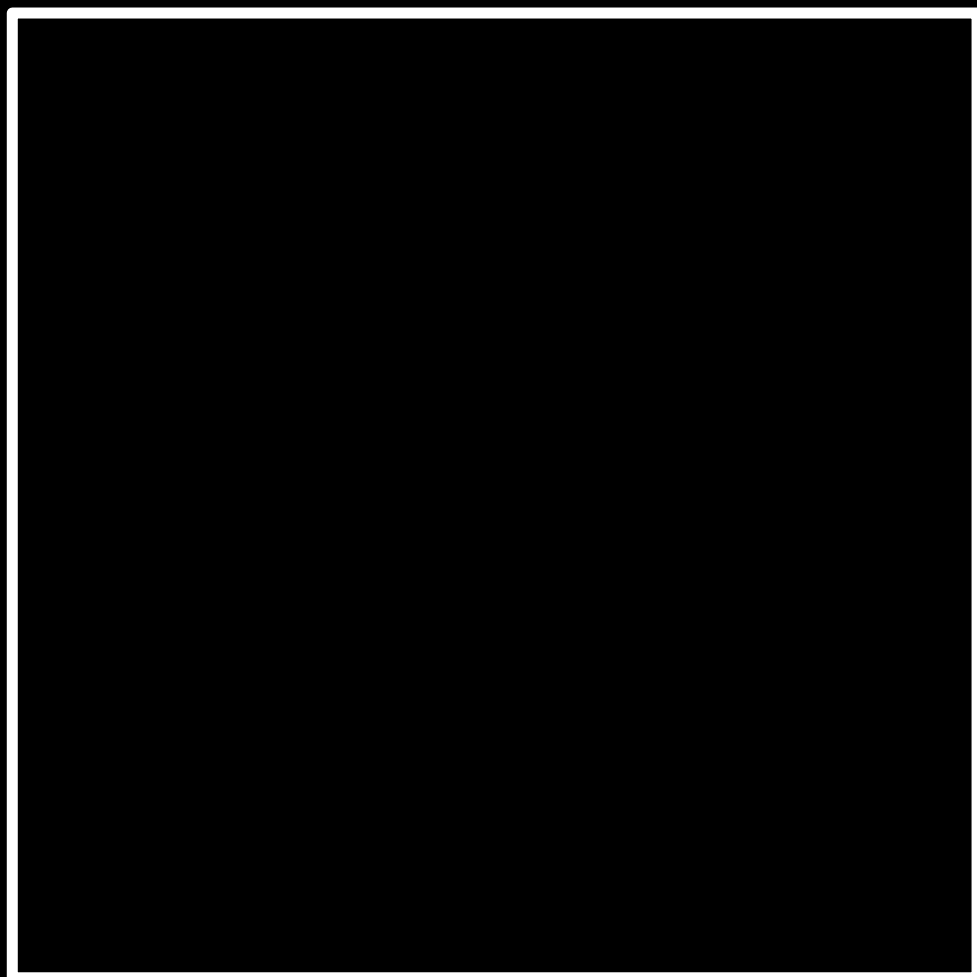


computer vision

computational methods for analyzing and
understanding digital images

[illegible]



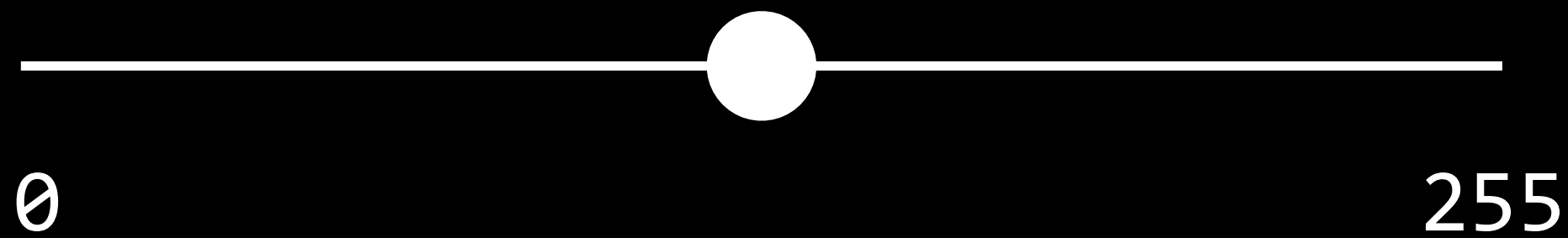
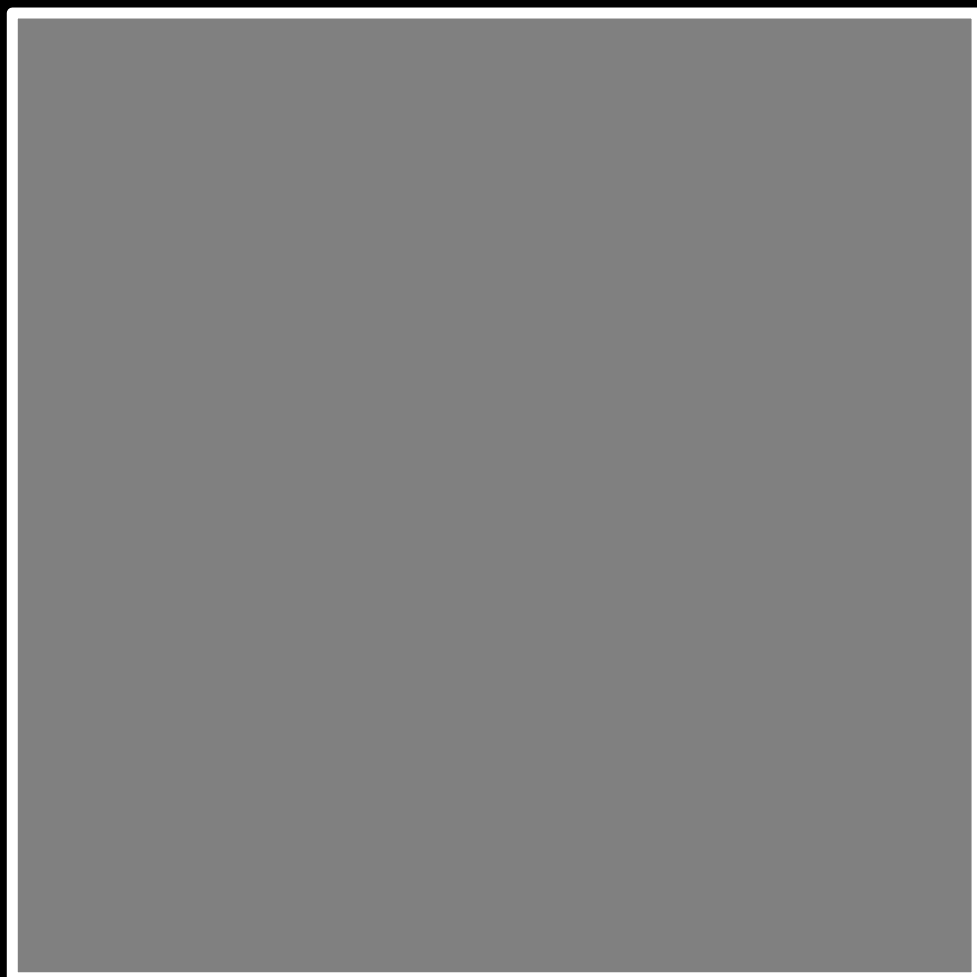


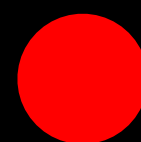
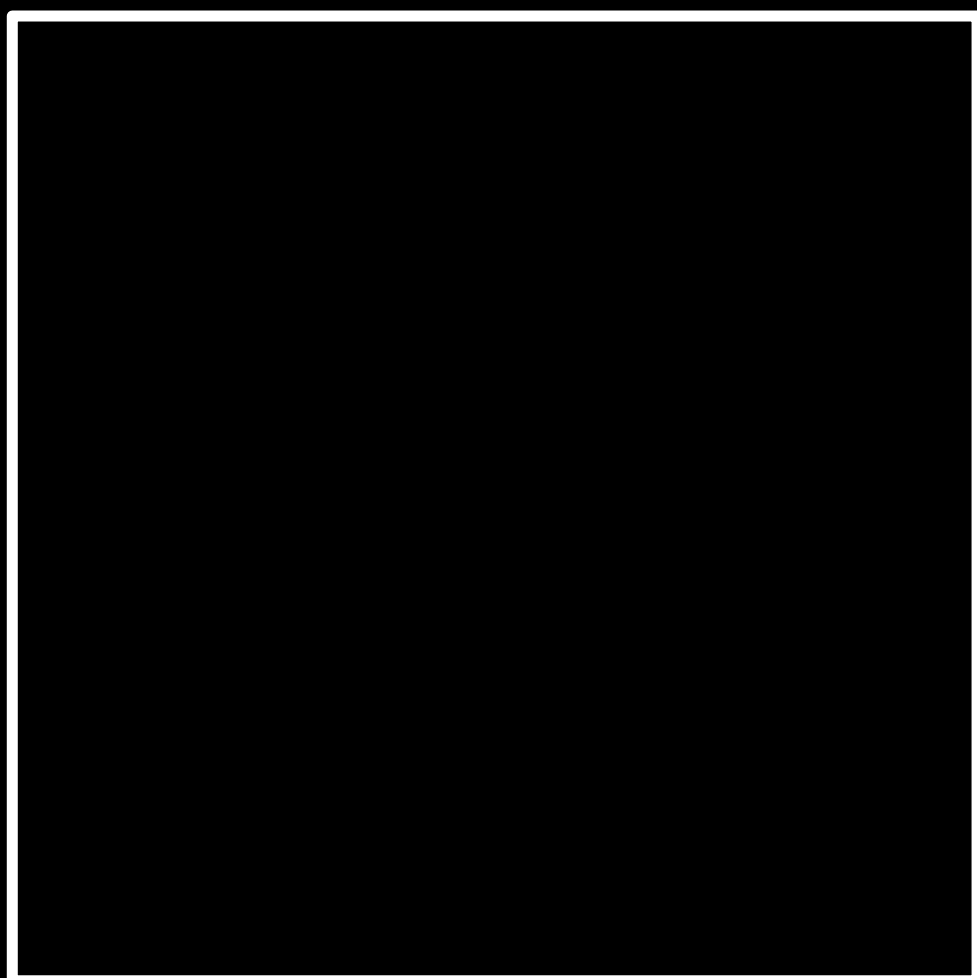
0



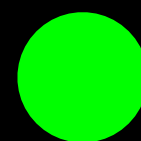
255



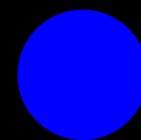




0



0



0



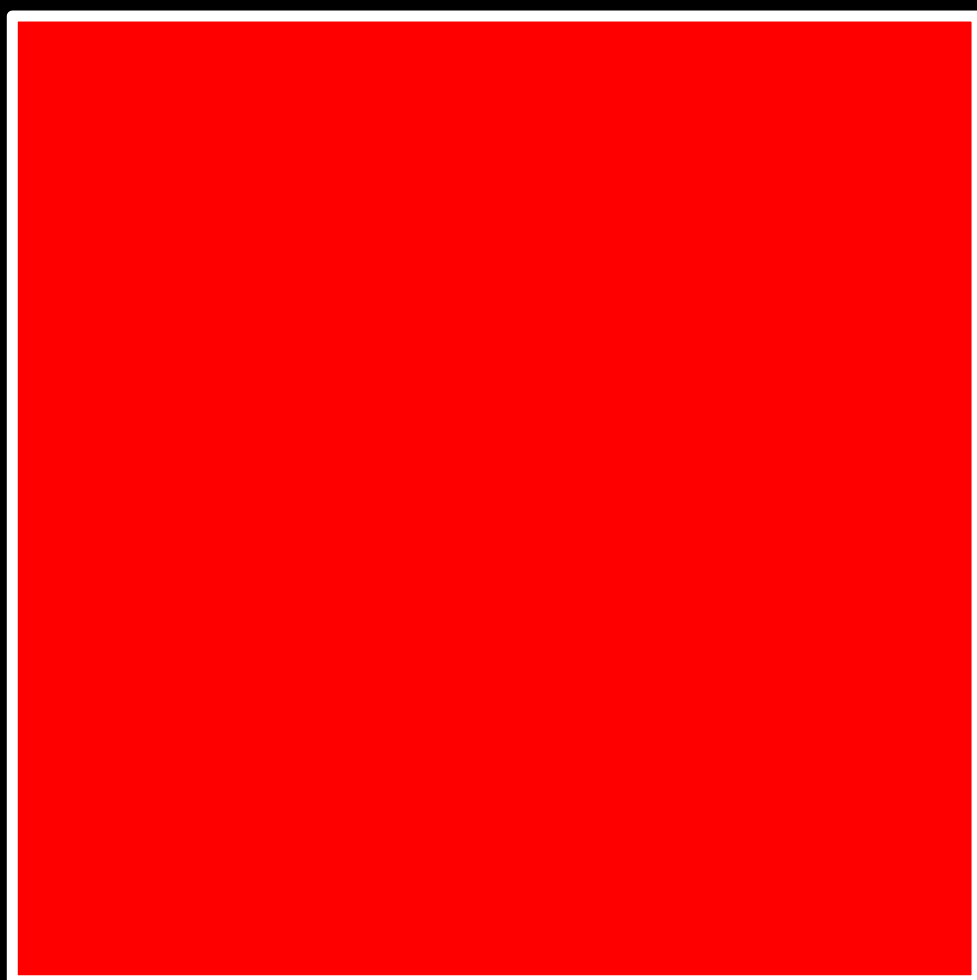
255

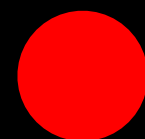
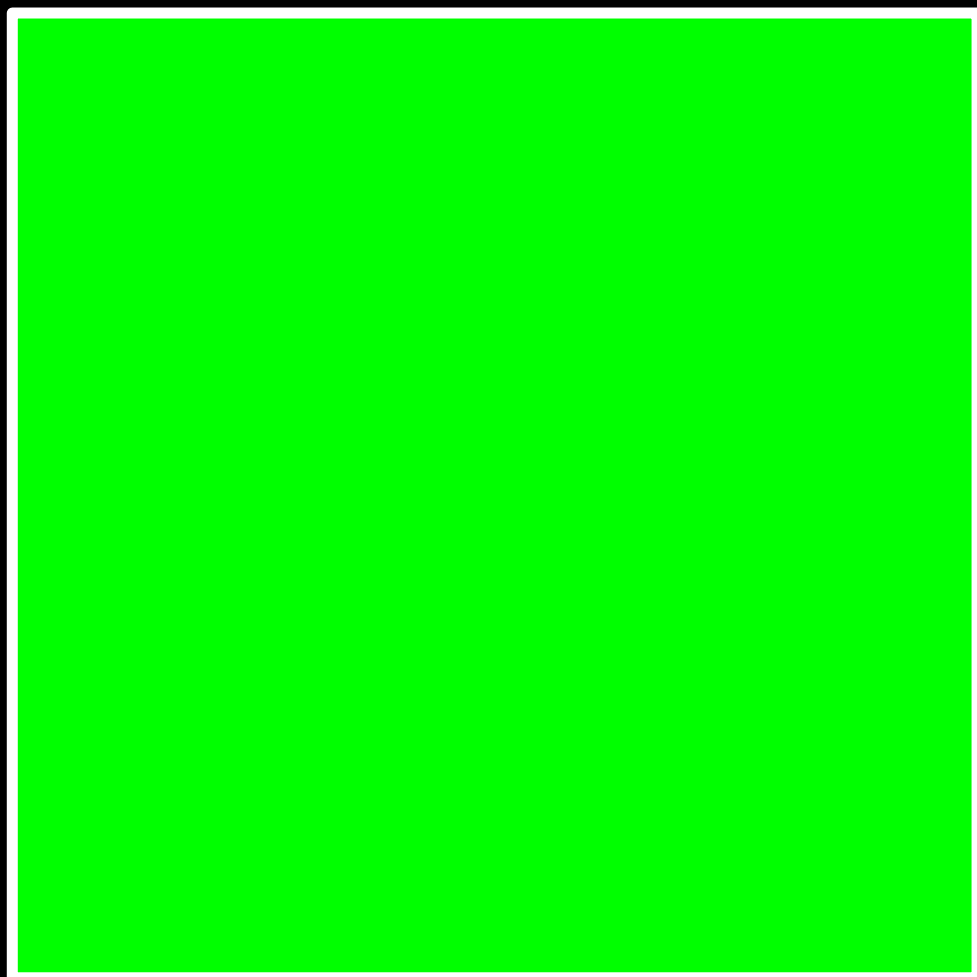


255



255





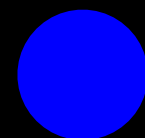
0

255



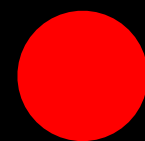
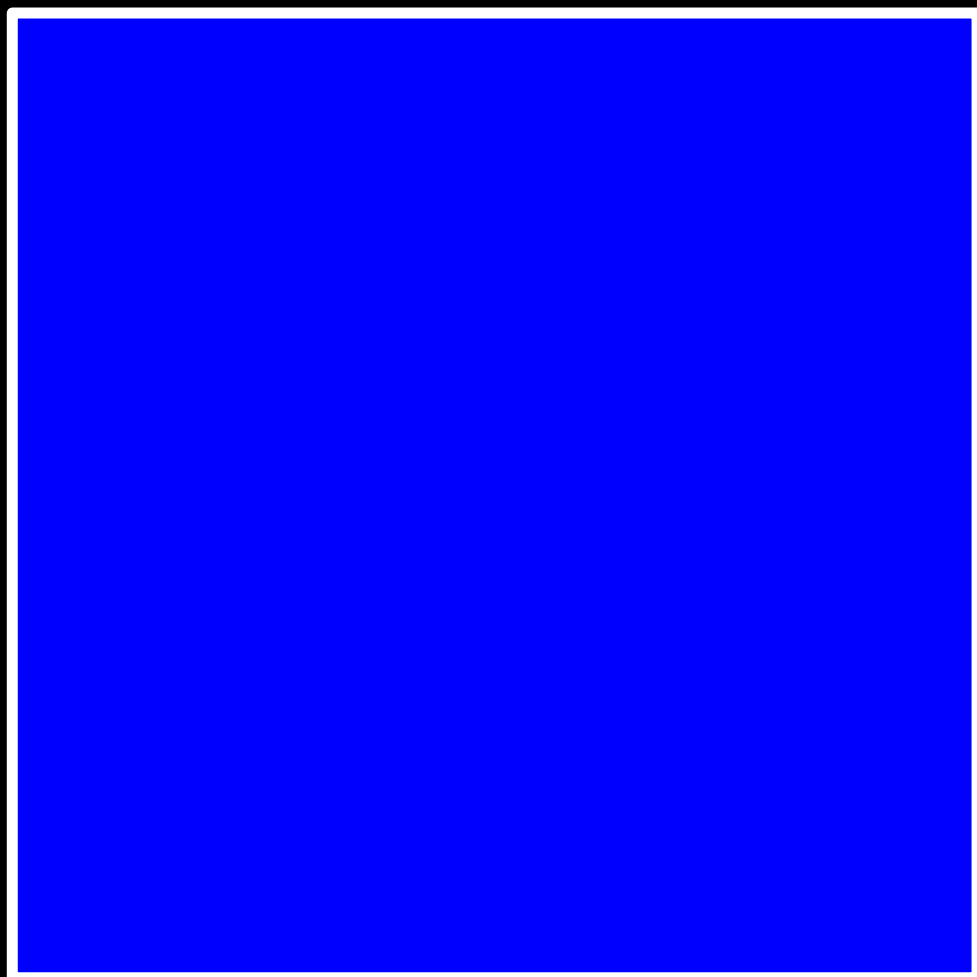
0

255



0

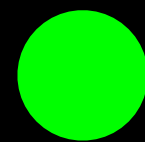
255



0



255



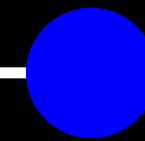
0



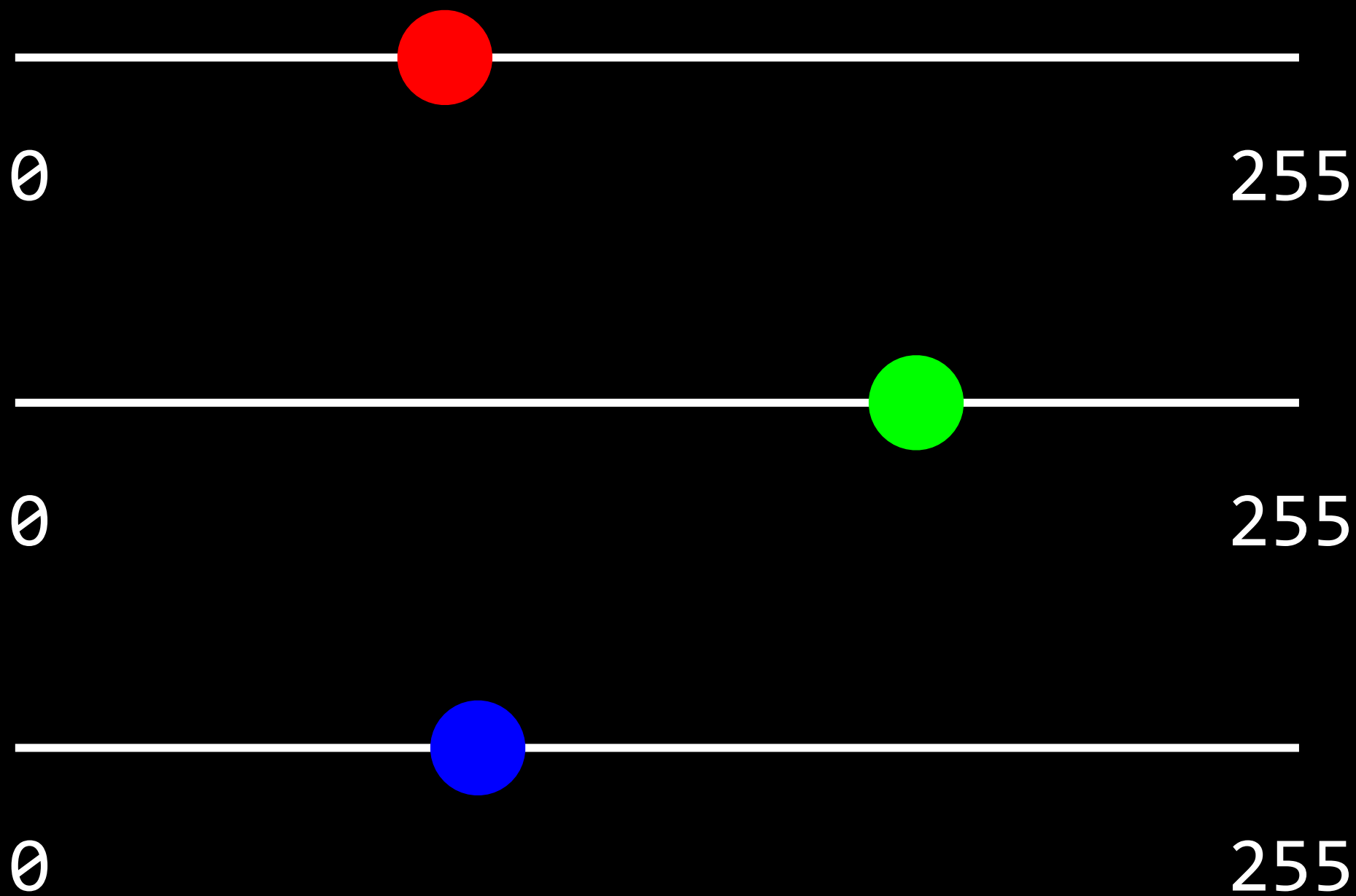
255



0



255



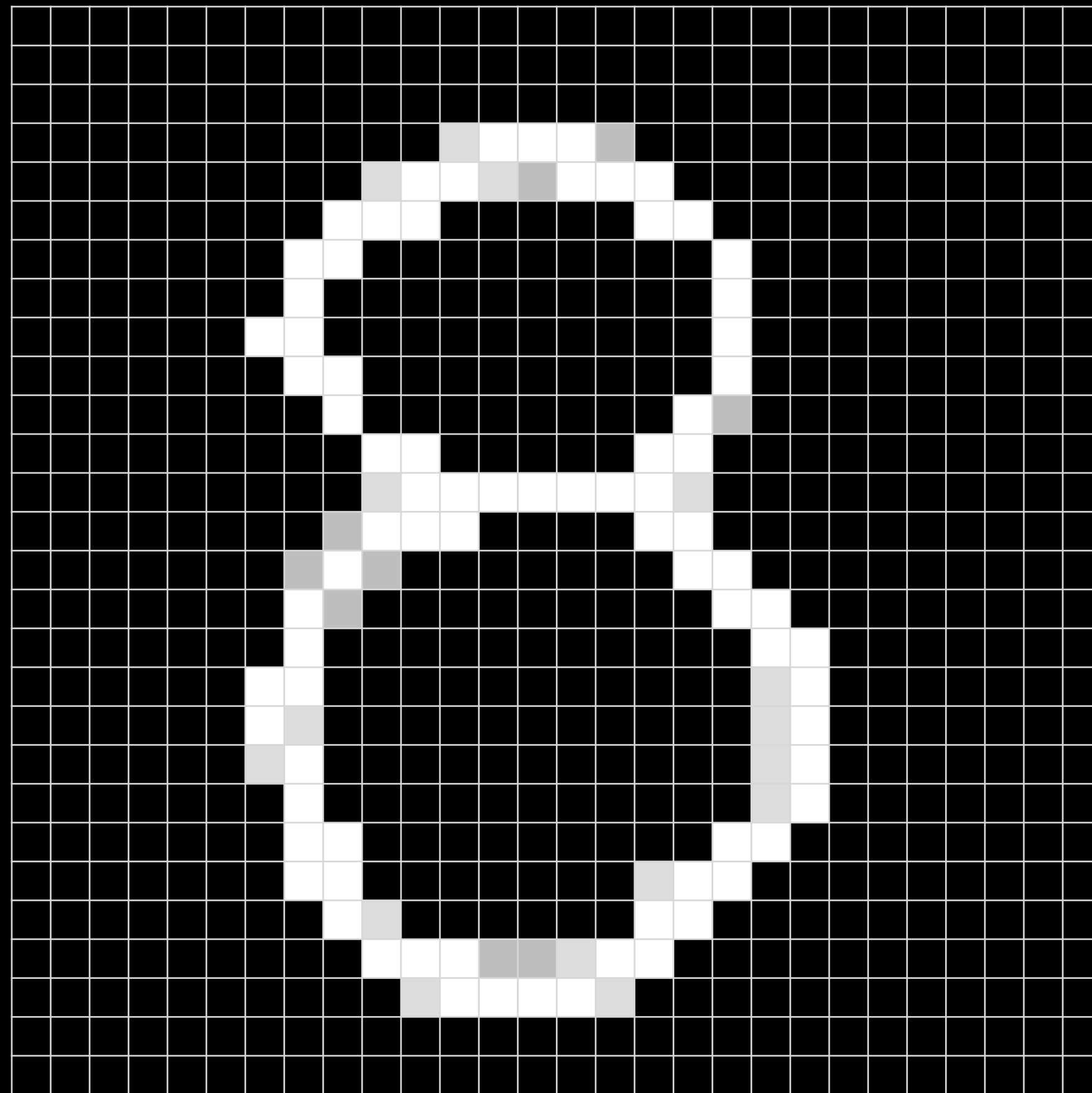
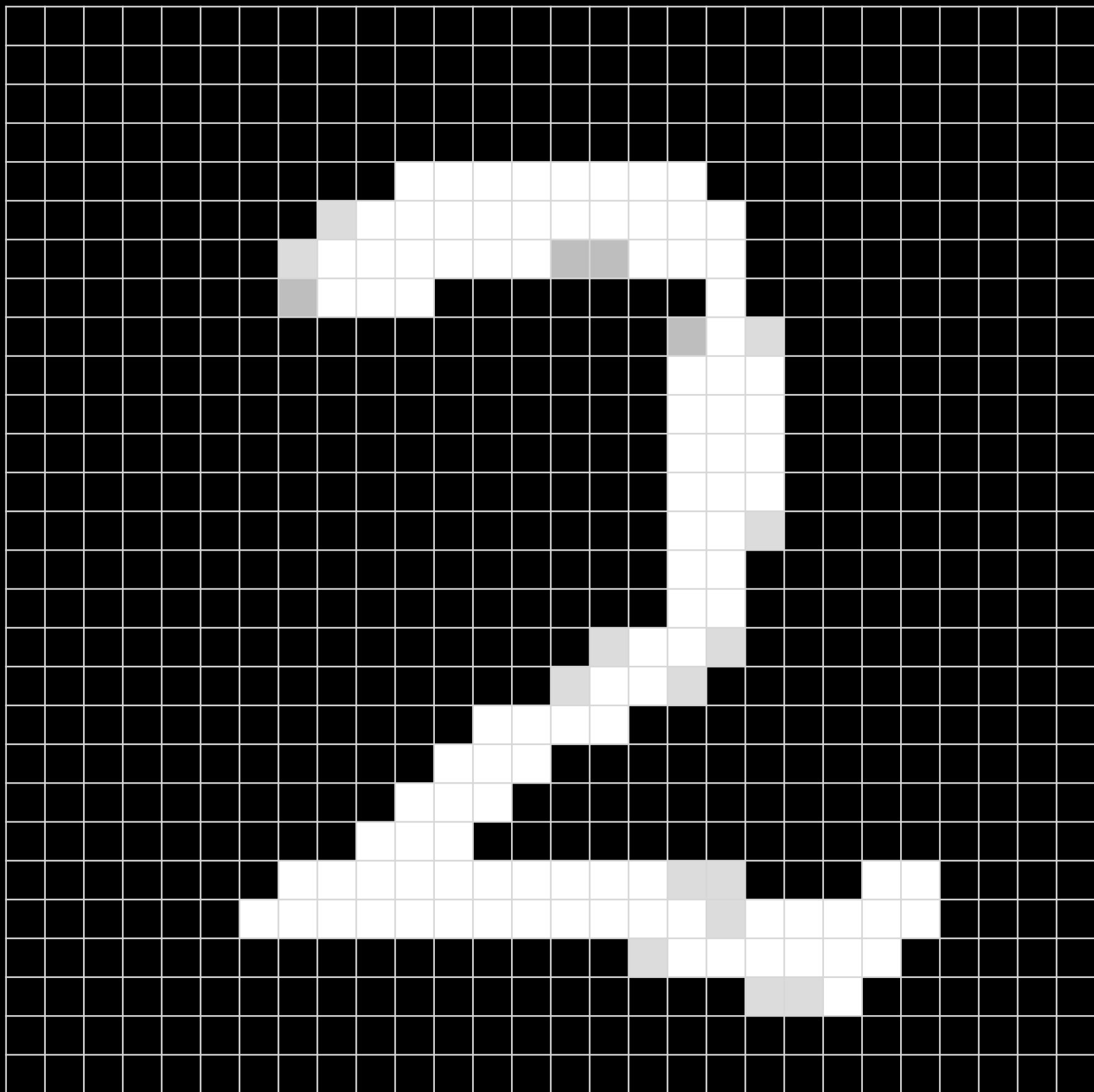


image convolution

applying a filter that adds each pixel value of an image to its neighbors, weighted according to a kernel matrix

0	-1	0
-1	5	-1
0	-1	0

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	20

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	20

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	20
40	

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	20
40	

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	20
40	50

10	20	30	40
10	20	30	40
20	30	40	50
20	30	40	50

0	-1	0
-1	5	-1
0	-1	0

10	20
40	50

-1	-1	-1
-1	8	-1
-1	-1	-1

20	20	20
20	20	20
20	20	20

-1	-1	-1
-1	8	-1
-1	-1	-1

$$\begin{aligned}
 &+ (20)(-1) + (20)(-1) + (20)(-1) \\
 &+ (20)(-1) + (20)(8) + (20)(-1) \\
 &+ (20)(-1) + (20)(-1) + (20)(-1)
 \end{aligned}$$

0

20	20	20
50	50	50
50	50	50

-1	-1	-1
-1	8	-1
-1	-1	-1

$$\begin{aligned}
 & (20)(-1) + (20)(-1) + (20)(-1) \\
 + & (50)(-1) + (50)(8) + (50)(-1) \\
 + & (50)(-1) + (50)(-1) + (50)(-1)
 \end{aligned}$$

90





pooling

reducing the size of an input by sampling
from regions in the input

max-pooling

pooling by choosing the maximum value in each region

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

50	

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

50	

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

50	110

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

50	110

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

50	110
20	

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

50	110
20	

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

50	110
20	40

30	40	80	90
20	50	100	110
0	10	20	30
10	20	40	30

50	110
20	40

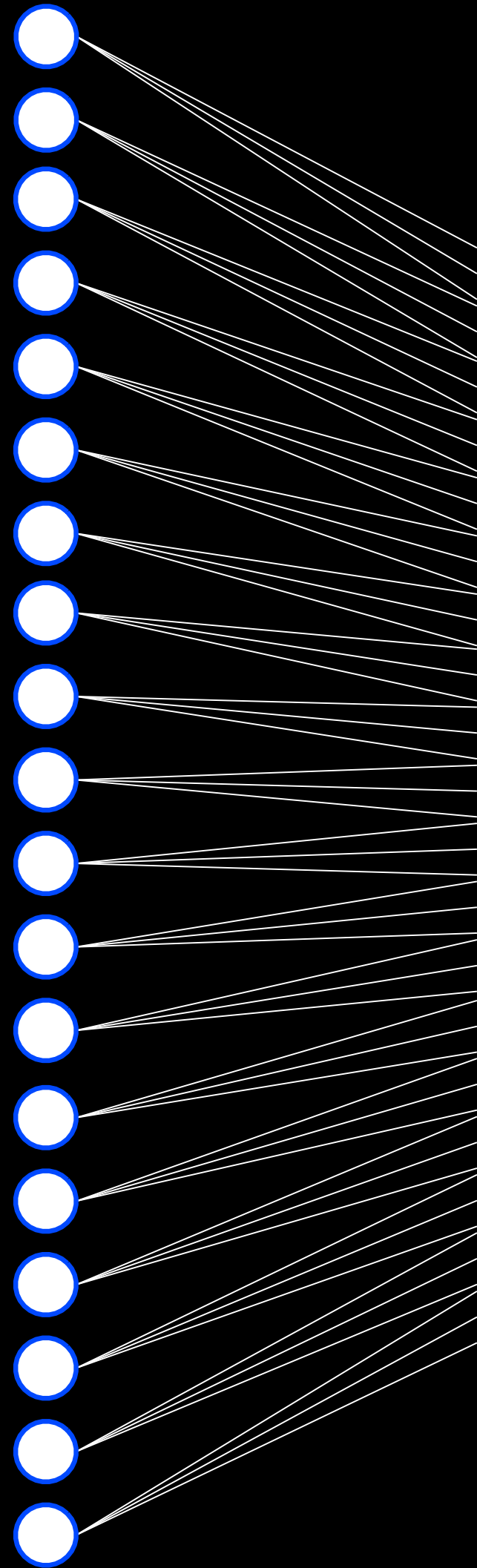
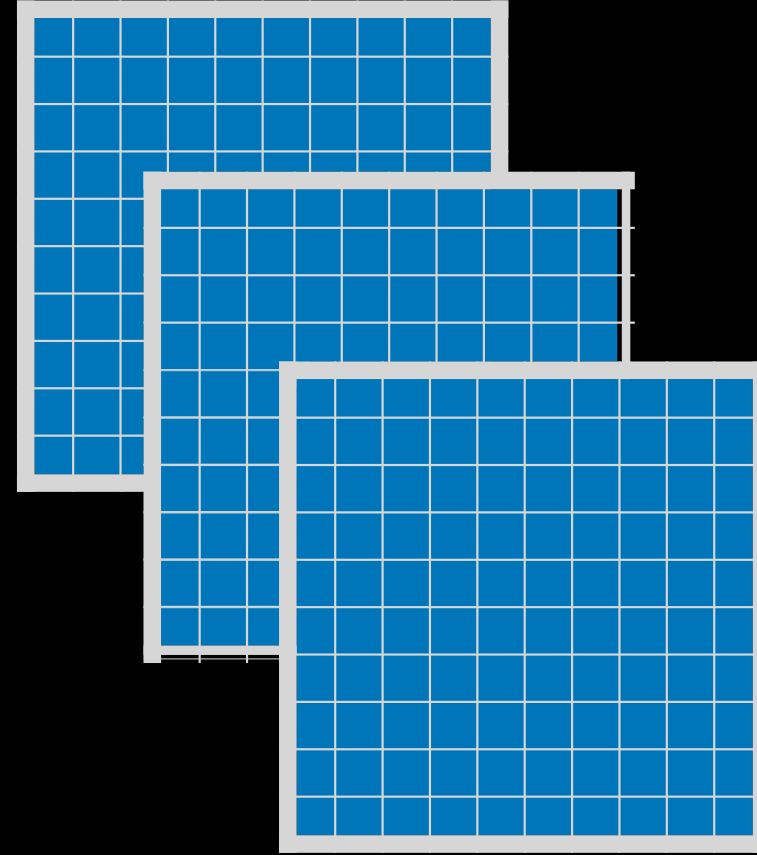
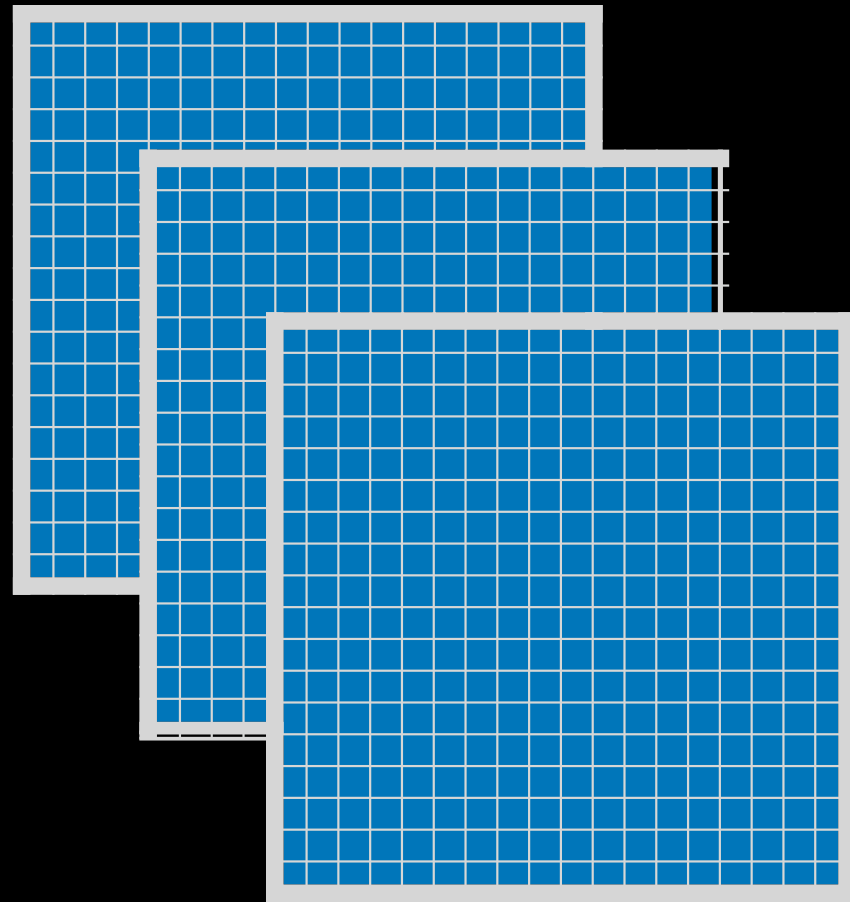
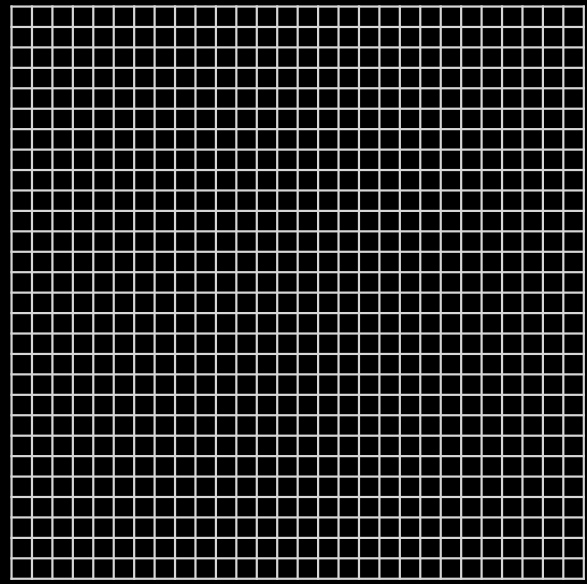
convolutional neural network

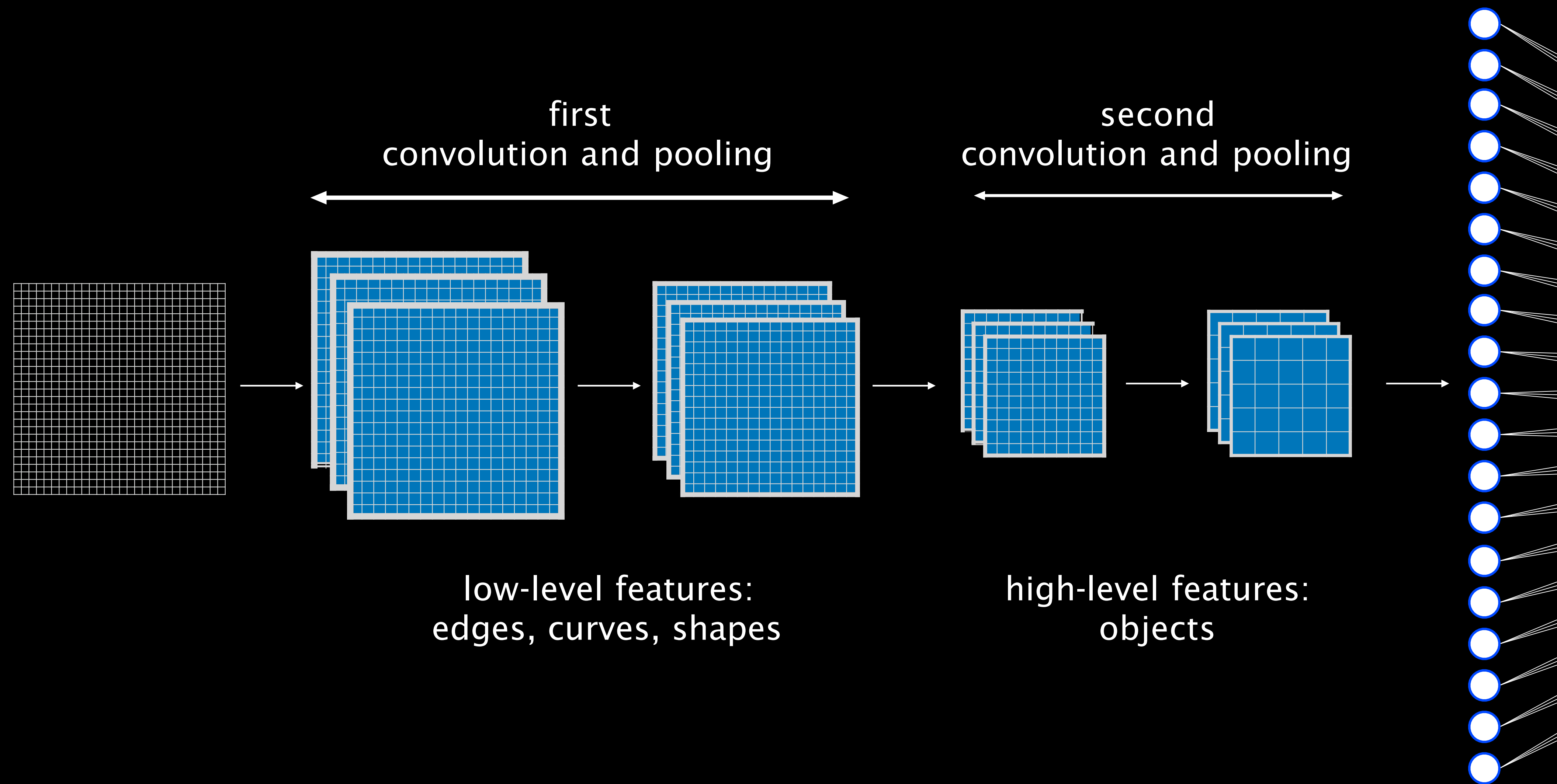
neural networks that use convolution,
usually for analyzing images

convolution

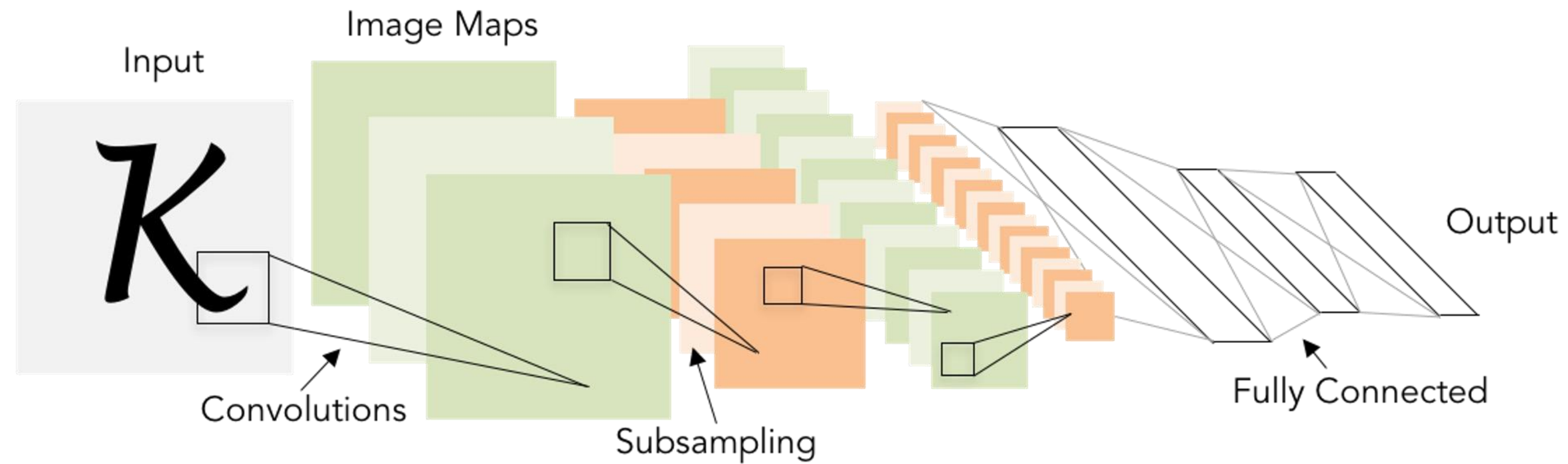
pooling

flattening





Convolutional Networks: LeCun et al, 1998



Applied backprop algorithm to a Neocognitron-like architecture

Learned to recognize handwritten digits

Was deployed in a commercial system by NEC, processed handwritten checks

Very similar to our modern convolutional networks!

IMAGENET Large Scale Visual Recognition Challenge

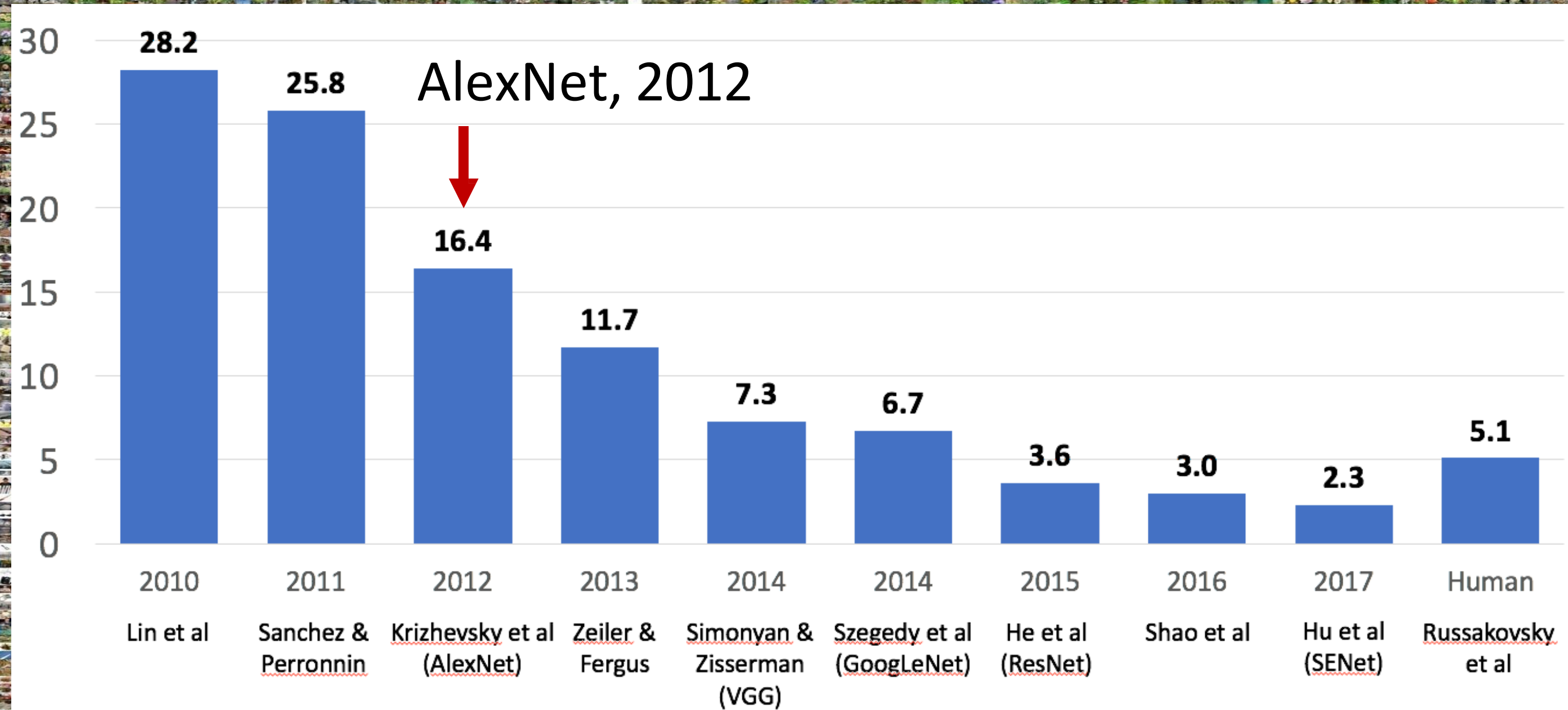
The Image Classification Challenge:
1,000 object classes
1,431,167 images



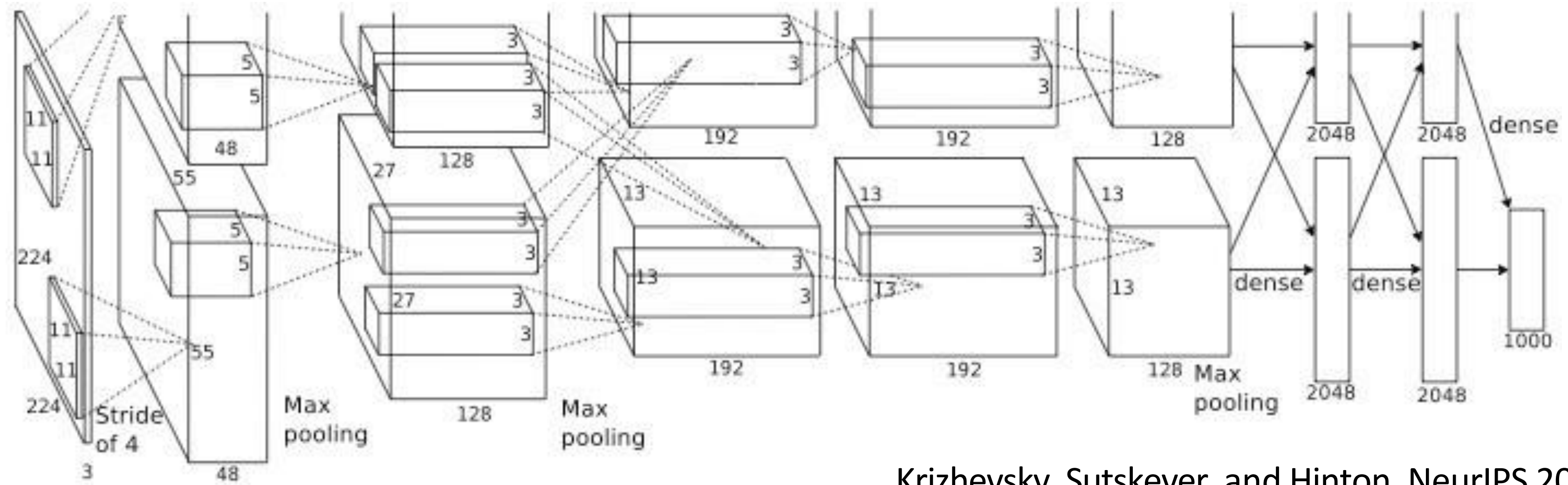
Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle

Deng et al, 2009
Russakovsky et al. IJCV 2015

IMAGENET Large Scale Visual Recognition Challenge



AlexNet: Deep Learning Goes Mainstream

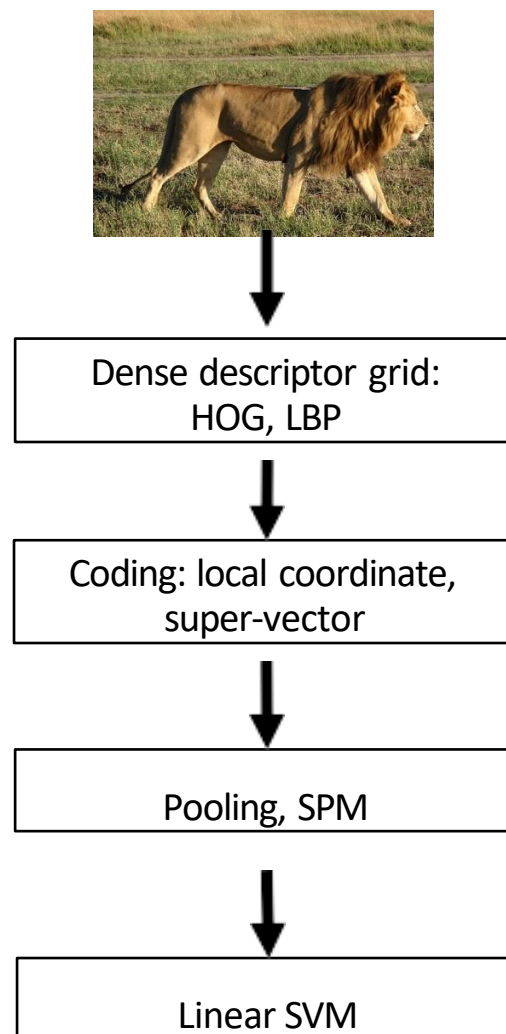


Krizhevsky, Sutskever, and Hinton, NeurIPS 2012

2010s

Year 2010

NEC-UIUC

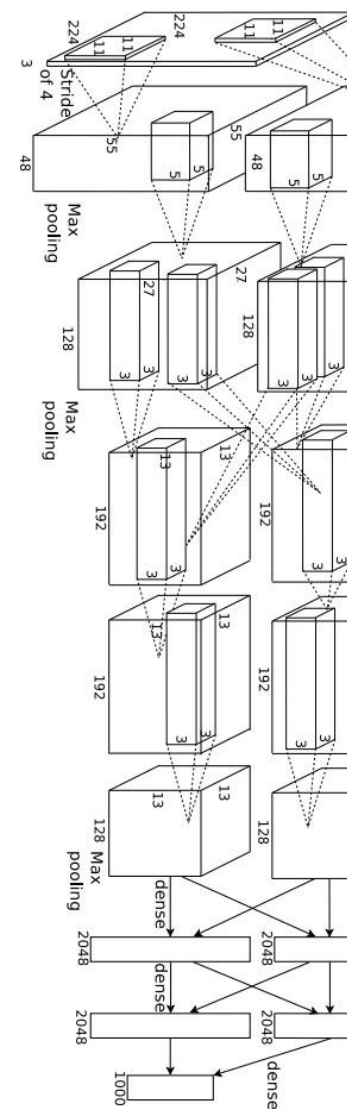


[Lin CVPR 2011]

[Lion image](#) by Swissfrog is licensed under CC BY 3.0

Year 2012

SuperVision

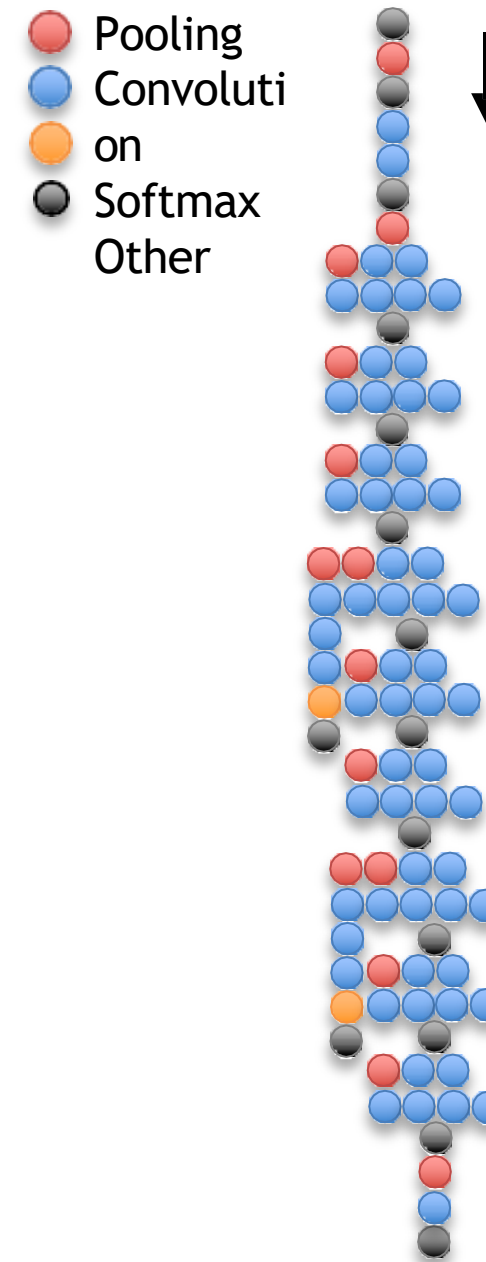


[Krizhevsky NIPS 2012]

Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

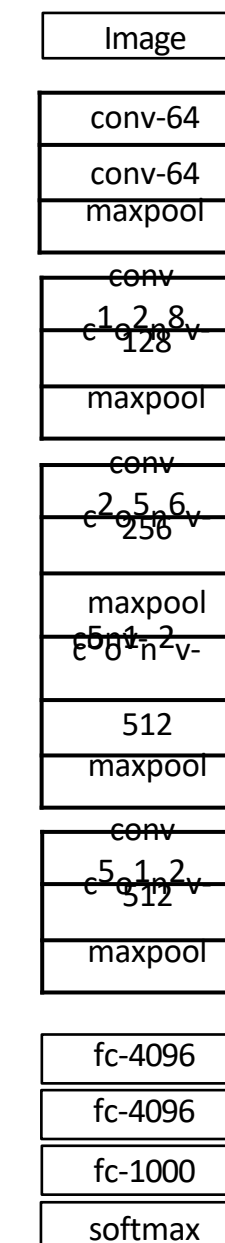
Year 2014

GoogLeNet



[Szegedy arxiv 2014]

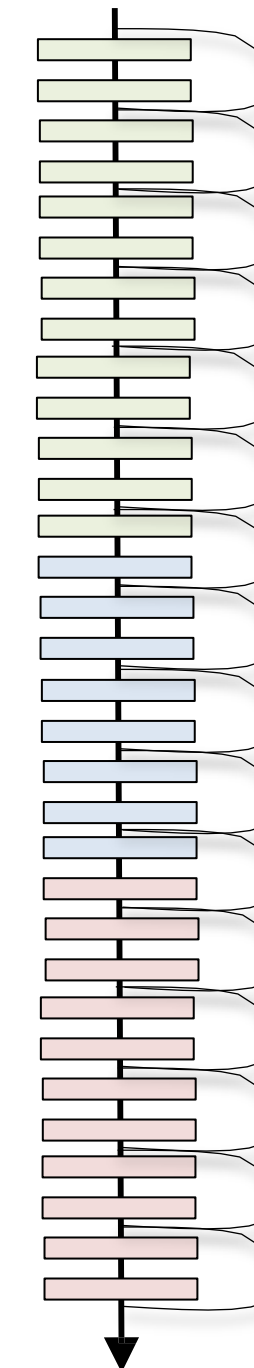
VGG



[Simonyan arxiv 2014]

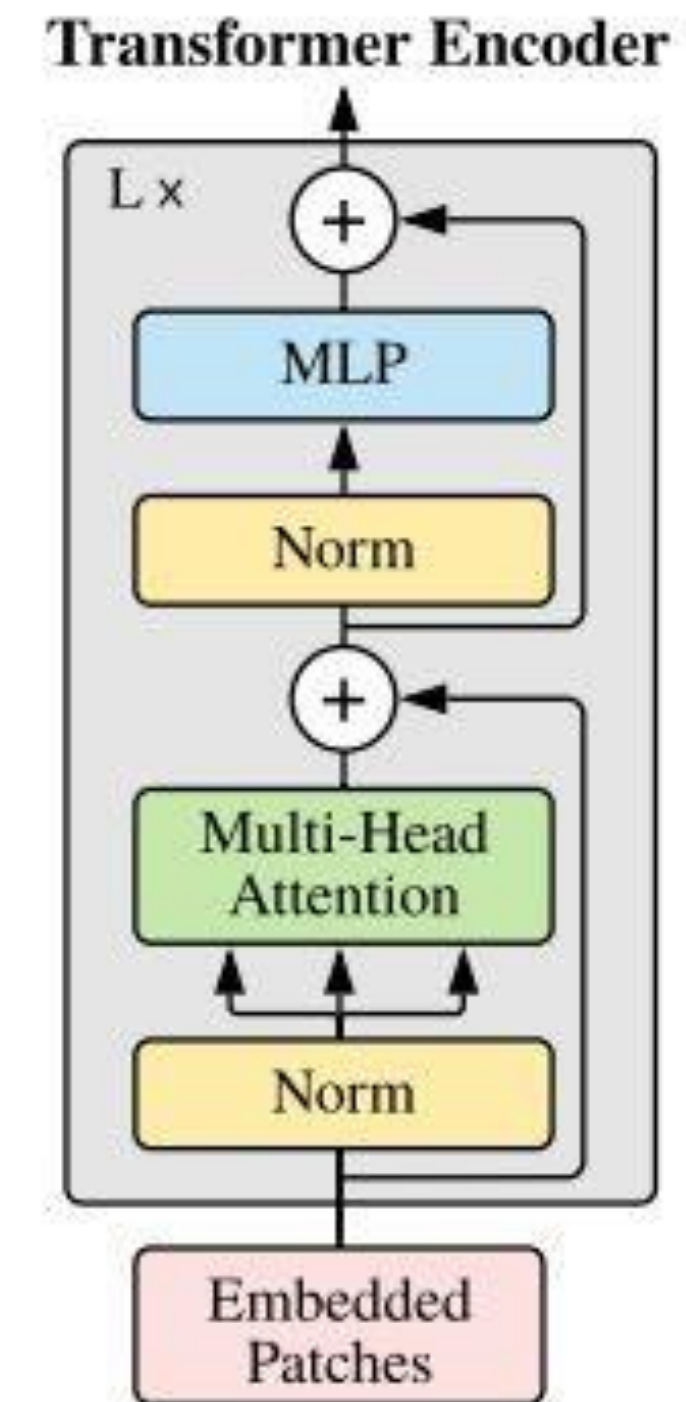
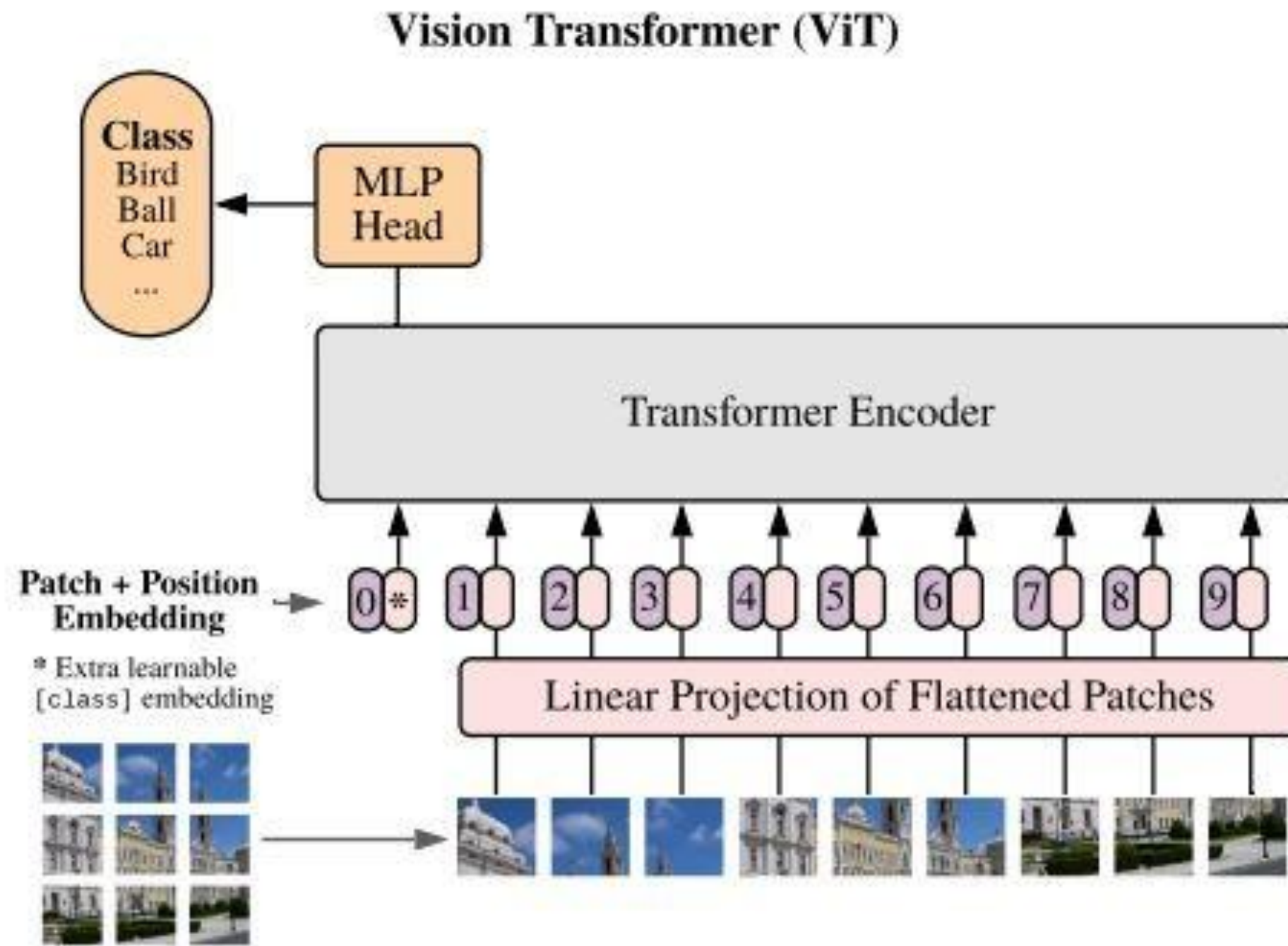
Year 2015

MSRA



[He ICCV 2015]

2020s



CV tasks

Image Classification

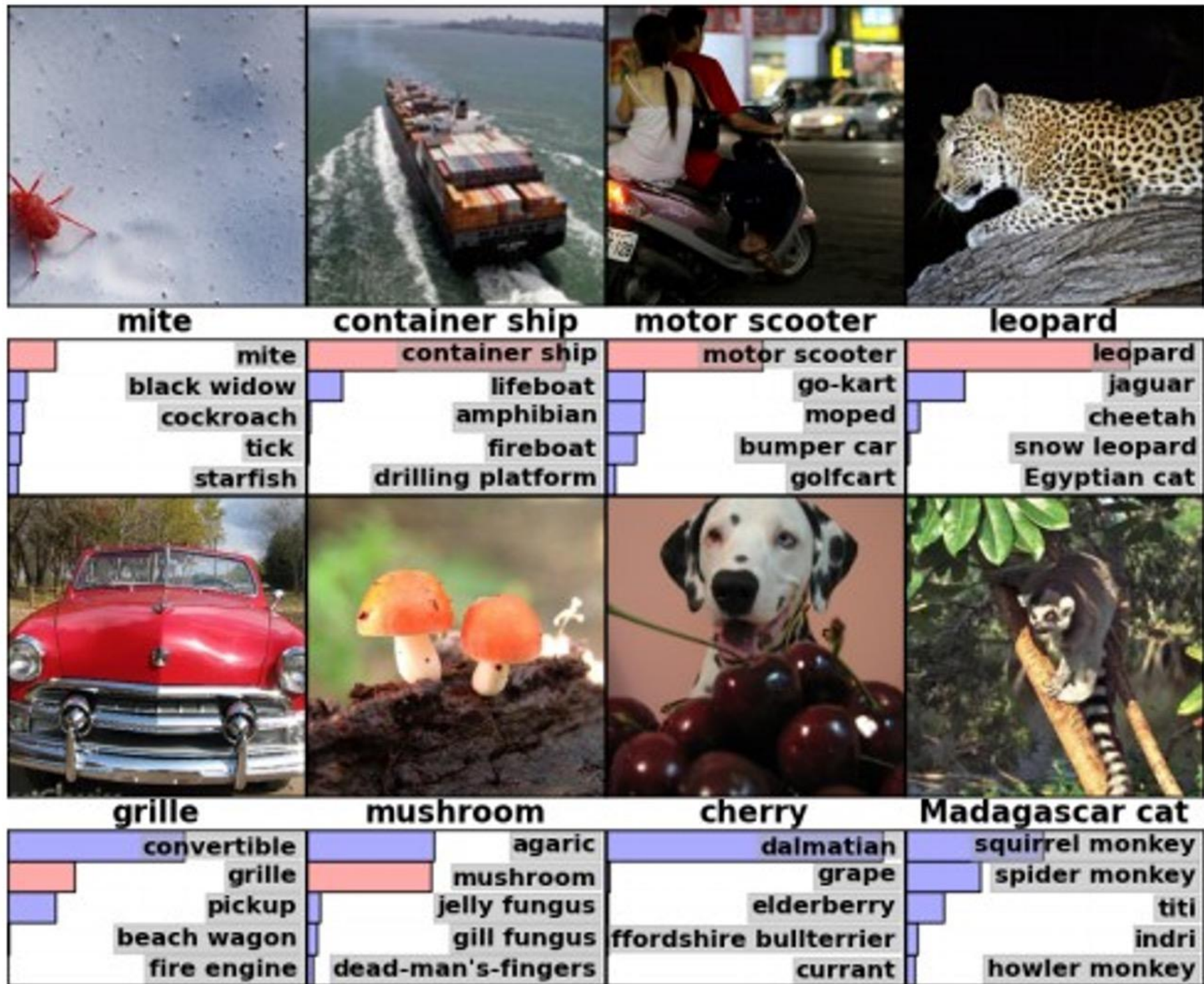
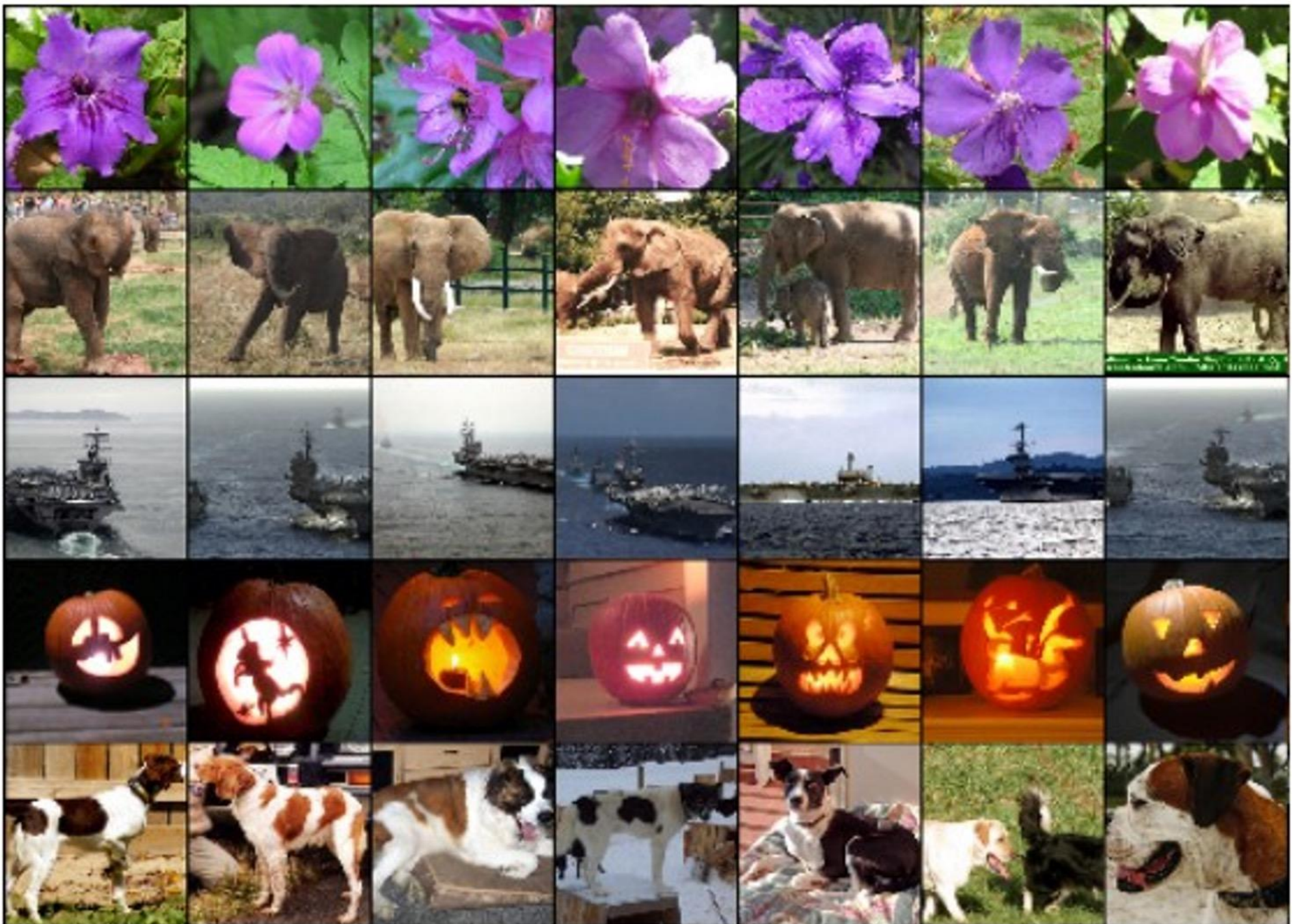


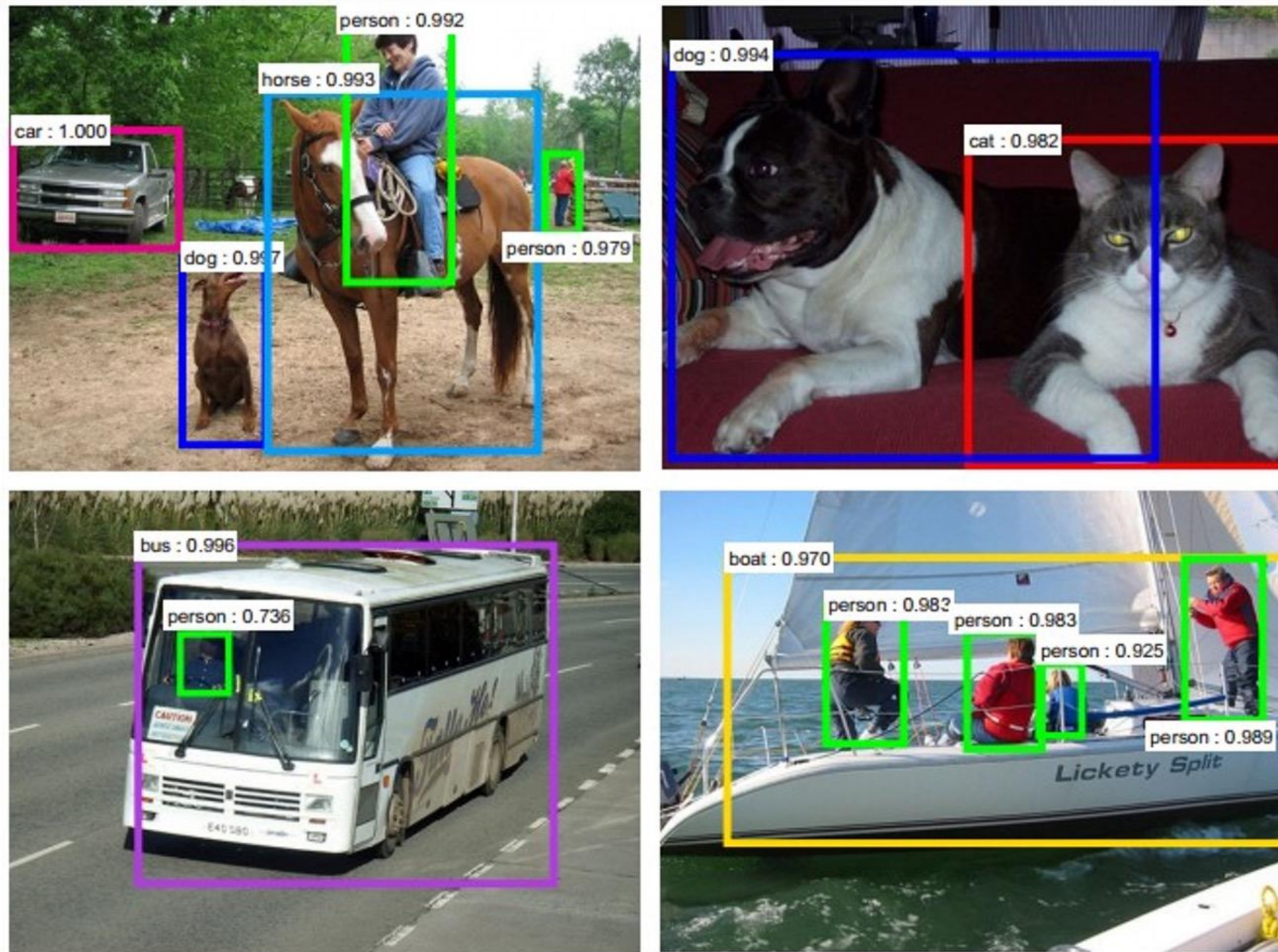
Image Retrieval



Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

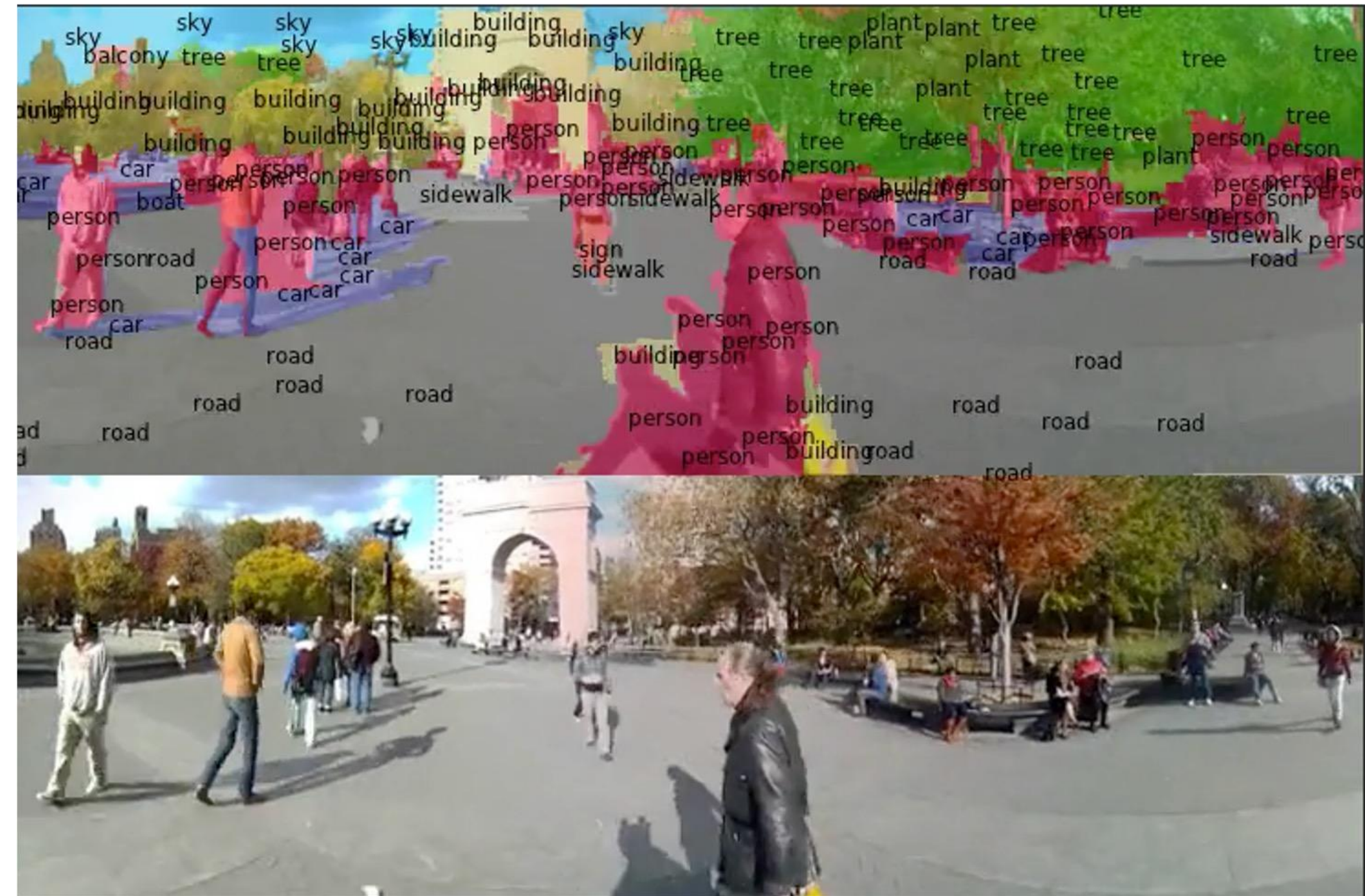
CV tasks

Object Detection



Ren, He, Girshick, and Sun, 2015

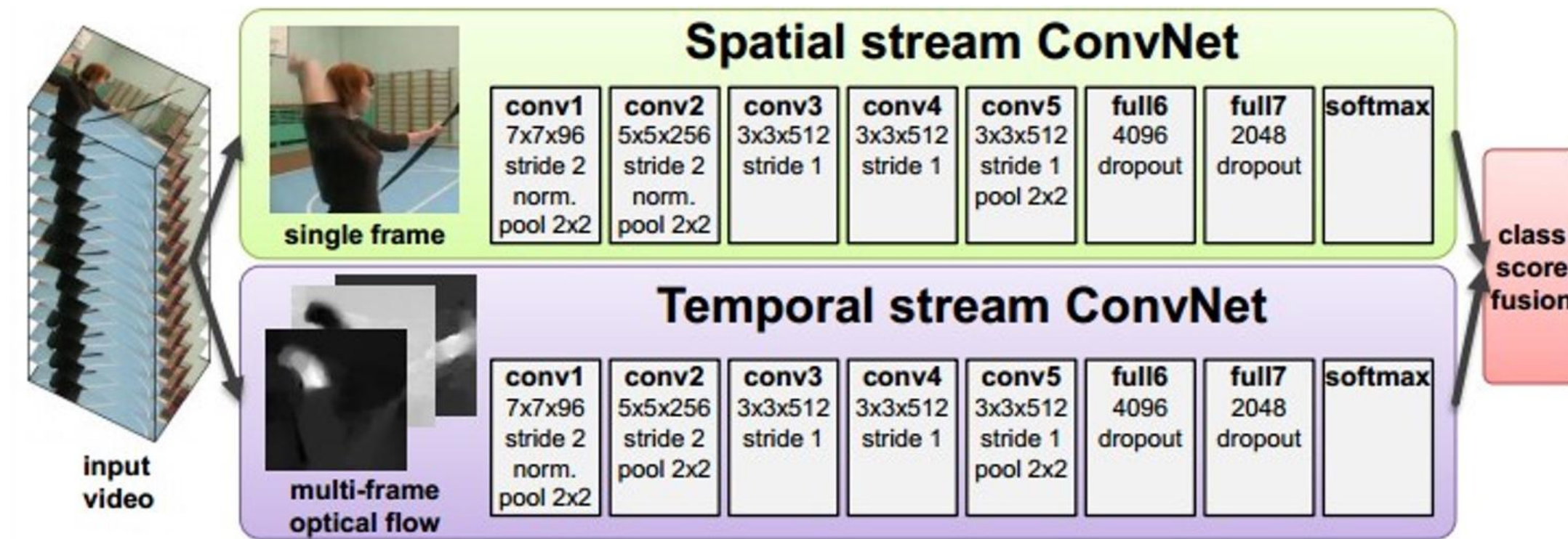
Image Segmentation



Fabaret et al, 2012

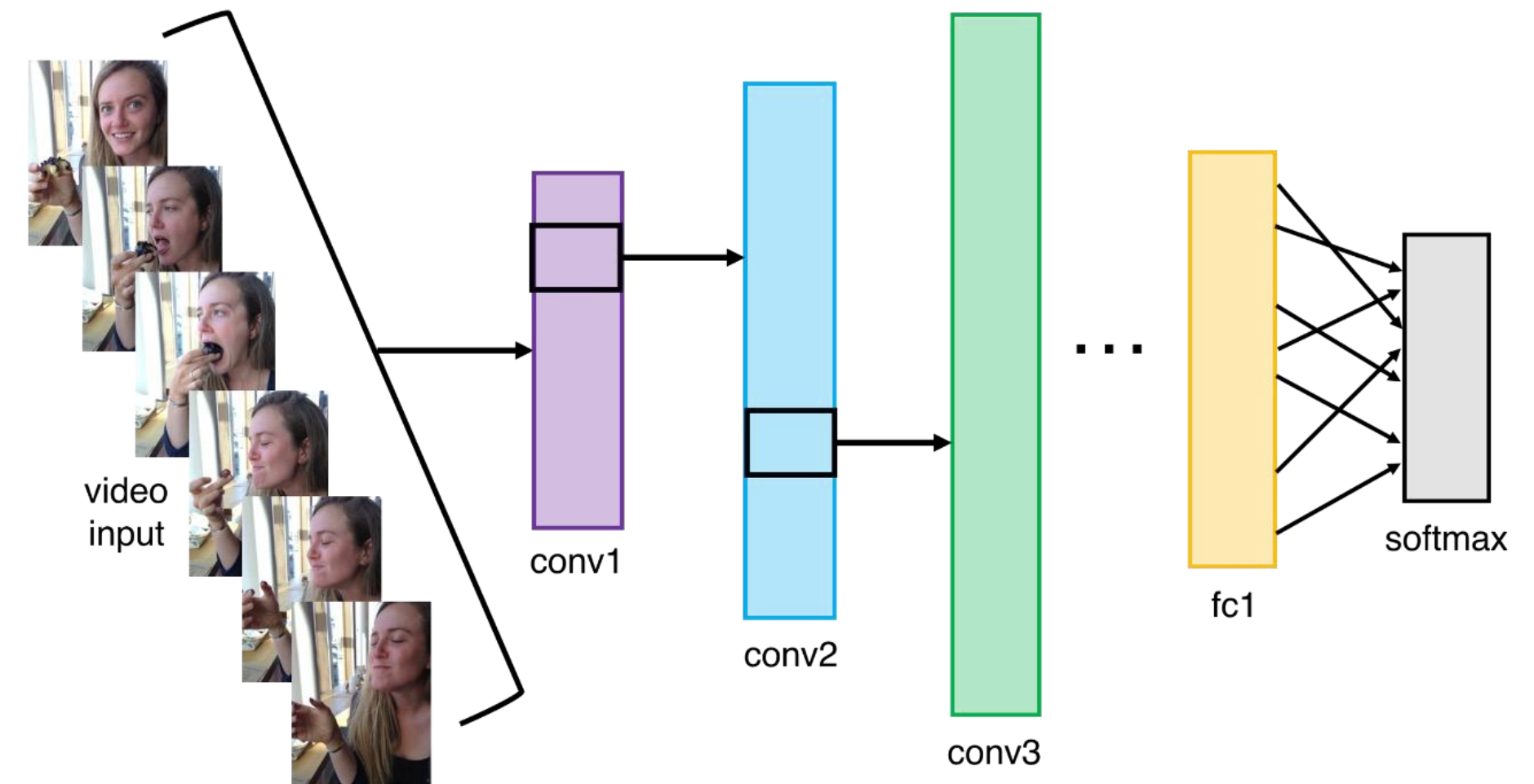
CV tasks

Video Classification



Simonyan et al, 2014

Activity Recognition



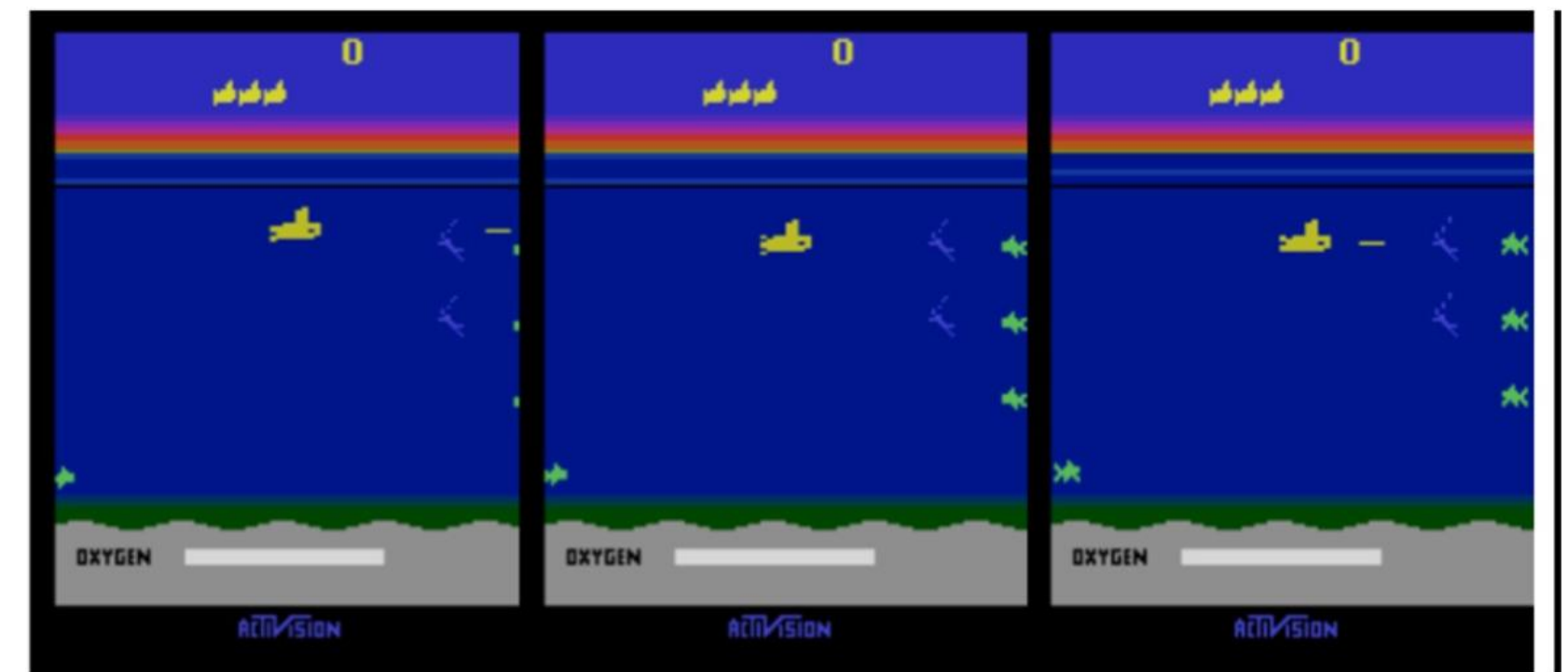
CV tasks

Pose Recognition (Toshev and Szegedy, 2014)



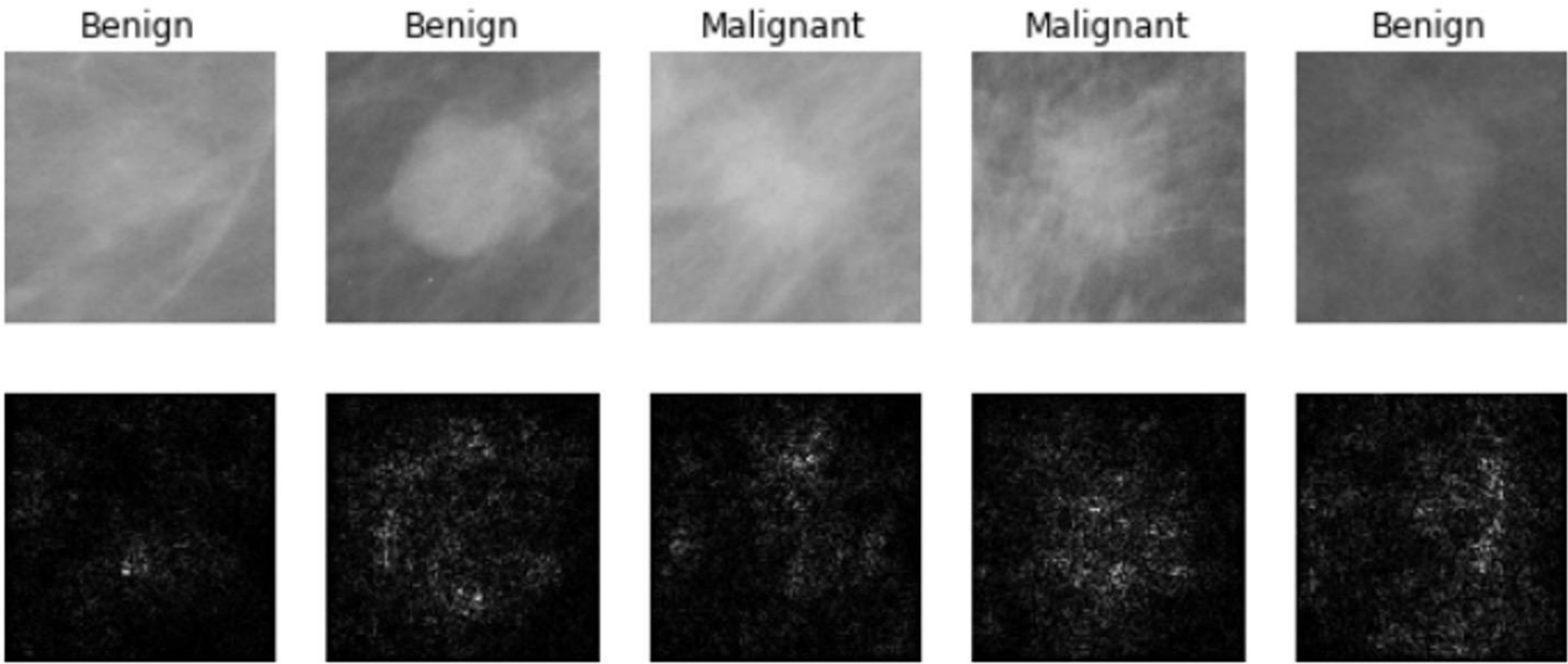
Playing Atari games (Guo et al, 2014)

frame:	t-3	t-2	t-1	t
"submarine"				
"diver"				
"enemy"				
"enemy+diver"				



CV tasks

Medical Imaging



Levy et al, 2016 Figure reproduced with permission

Galaxy Classification



Dieleman et al, 2014

From left to right: [public domain by NASA](#), usage [permitted by ESA/Hubble](#), [public domain by NASA](#), and [public domain](#).

Whale recognition



Kaggle Challenge

[This image](#) by Christin Khan is in the public domain and originally came from the U.S. NOAA.

CV tasks

Image Captioning

Vinyals et al, 2015

Karpathy and Fei-Fei, 2015



*A white teddy bear
sitting in the grass*



*A man in a baseball
uniform throwing a ball*



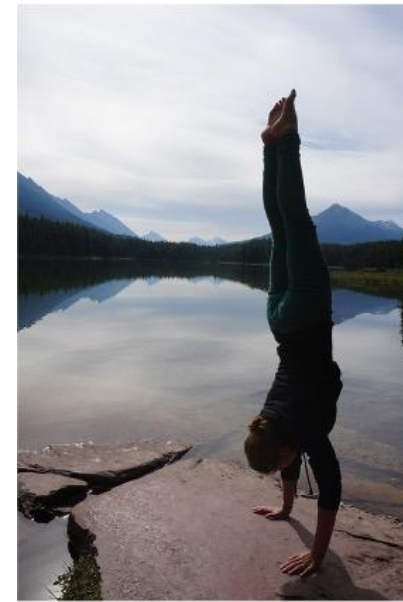
*A woman is holding
a cat in her hand*



*A man riding a wave
on top of a surfboard*



*A cat sitting on a
suitcase on the floor*



*A woman standing on a
beach holding a surfboard*

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<https://pixabay.com/en/teddy-plush-bears-cute-teddy-bear-1623436/>
<https://pixabay.com/en/surf-wave-summer-sport-litoral-1668716/>
<https://pixabay.com/en/woman-female-model-portrait-adult-983967/>
<https://pixabay.com/en/handstand-lake-meditation-496008/>
<https://pixabay.com/en/baseball-player-shortstop-infield-1045263/>

Captions generated by Justin Johnson using [NeuralTalk2](#)



[Original image](#) is CC0 public domain
[Starry Night](#) and [Tree Roots](#) by Van Gogh are in the public domain
[Bokeh image](#) is in the public domain
Stylized images copyright Justin Johnson, 2017;
reproduced with permission

Mordvinsev et al, 2015
Gatys et al, 2016

Slide inspiration: Justin Johnson

CV tasks

TEXT PROMPT

an armchair in the shape of an avocado. an armchair imitating an avocado.

AI-GENERATED IMAGES

